

Choline- and betaine-defined diets for use in clinical research and for the management of trimethylaminuria.

This article describes the development of a series of choline- and betaine-controlled diets that were served to research subjects as part of an ongoing study of diet requirements in humans. These diets were developed based on the analysis of choline and betaine in individual foods. The calculated diets were compared with analyses of all foods combined into a single sample for each day. The laboratory analyses of choline and betaine in the whole-diet aliquots matched the estimated amounts in the diets that were calculated from the analyses of individual foods. These diets were adjusted for several levels of choline and betaine and were well accepted by research subjects who consumed them for a time period of up to 2 months. This article describes applications of this diet for use in clinical research on methyl-group requirements in humans and for use in clinical practice for counseling the client who requires a choline-controlled diet.